📘 CHEMICAL BONDING (ADV → OLYMPIAD LEVEL)

1. Ionic vs Covalent – REAL LOGIC

Not pure ionic/covalent

→ always partial character

Fajan Rule (deep understanding)

Covalent character ↑ when:

cation small + high charge

anion large

👉 Example: $AlCl_3$ covalent, NaCl ionic

2. Lattice Energy (IMPORTANT)

Depends on:

charge ↑ → lattice energy ↑

size ↓ → lattice energy ↑

👉 Use:

stability of ionic compounds

melting point

3. VSEPR Theory – LIMITATION + REALITY

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Predicts shape but:

lone pair-lone pair repulsion strongest

order: LP-LP > LP-BP > BP-BP

👉 distortion hota hai actual geometry me

4. Hybridization (ADV THINKING)

Steric number = sigma bonds + lone pairs

Important twist:

Lone pair compress angle

More s-character → shorter bond

5. Bent's Rule (OLYMPIAD 🔥)

Atomic orbitals hybridization adjust karta hai:

"More electronegative atom → more p-character bond"

👉 Result:

electronegative atom ke bond longer hote hain

6. Molecular Orbital Theory (MOT) – DEEP

Bond order = (Nb - Na) / 2

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Stability:

BO > 0 → stable

7. MOT Energy Order (CRITICAL TRAP)

For B2, C2, N2: pi2p < sigma2p

For O2, F2: sigma2p < pi2p

👉 Reason: s-p mixing

8. Magnetic Nature

Unpaired electrons → paramagnetic

👉 Example: O2 paramagnetic (VERY IMPORTANT)

9. Hydrogen Bonding (ADV)

Types:

Intermolecular → increases BP

Intramolecular → decreases BP

👉 Intramolecular → chelation type stabilization

10. Back Bonding (OLYMPIAD GOLD 🔥)

Definition: Lone pair donation from filled orbital to empty orbital

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Example: BF3

👉 Effect:

bond length ↓

stability ↑

11. Dipole Moment (ADV)

mu = charge × distance

Key insight:

symmetric molecule → zero dipole

Example: CO2 = 0

H2O ≠ 0

🔥 FINAL CHEMICAL BONDING INSIGHT

Bonding = electron distribution + energy minimization

Real questions = multiple concepts combine

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